

Master's final year internship report

Reconstructing comparative atlases of fish
genomes, to better understand evolutionary
trajectories across species.

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Finally, I would like to thank everyone who helped me in any way during this period.

List of abbreviations

- 1R:** First Round (first genomic duplication)
- 2R:** Second Round (second genomic duplication)
- 3R:** Third Round (third genomic duplication)
- 4R:** Fourth Round (fourth genomic duplication)
- CAR:** Chromosomal Ancestral Region
- Cs4R:** Cyprinid-specific 4th Round
- FAIR:** Findable, Accessible, Interoperable, Reusable
- GFF:** General Feature Format
- LPGP:** Laboratoire de Physiologie et Génomique des Poissons
- NCBI:** National Center for Biotechnology Information
- RNA-seq:** RNA sequencing
- SCORPiOs:** Synteny-guided Correction of Orthologs
- Ss4R:** Salmonid-specific 4th Round
- TPM:** Transcripts Per Million
- Ts3R:** Teleost-specific 3rd Round
- WGD:** Whole-Genome Duplication

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I. Introduction

Biological evolution is one of the oldest fields of study in the life sciences, dealing with many other scientific disciplines such as geology, ethology and informatics. It enables us to answer important questions about human, vertebrate and plant evolution, as well as the evolution of life from the first primary organism to the most evolved. The technological revolution has recently increased our knowledge of the biological mechanisms of evolution and opened new areas of research.

A. Evolution of vertebrates across time

I.A.1. Study of evolution – history, from 19th century to nowadays

The term 'natural selection' was first used by Charles Darwin and Alfred Russel Wallace in the nineteenth century. Natural selection is based on the idea that individuals are born with a set of attributes. If they are adapted to their environment, they will survive and pass on their adaptations to the next generation. Those who are not adapted will not survive, thus we can talk about “the struggle for life”¹.

At the same time, between 1856 and 1863, Gregor Mendel, the “father of modern genetics”, was conducting experiments on plants in the garden of his monastery. He published his findings in 1866, in which he discussed Mendelian inheritance, a form of biological inheritance. These could be summarised as “Mendel’s three laws of inheritance”: The law of dominance and uniformity; the law of segregation; and the law of independent assortment. Unfortunately, his work was received as papers on hybridisation rather than inheritance. This lack of recognition of Mendel's work explains why Darwin struggled to explain the genetic mechanism: he was simply unaware of it, even though they were working during the same period¹. The importance of this principle was recognised by scientists in the early 20th century. Darwin's and Wallace's theory was combined with Mendelian genetics in the 1930s and 1940s, forming the basis of the modern synthesis of evolutionary biology¹.

Then, in the 20th century, there was a succession of brilliant evolutionary biologists with their new theories and refinements of old concepts, until the development of computer technology. Combined with mathematics, this allowed researchers to create tools to model organismic evolution¹. More recently, since the beginning of the twenty-first century, improvements in genomic techniques, the sequencing revolution and the emergence of powerful new technologies (not to mention the current growth in AI usage) have ushered in a new era for almost all biological fields. For example, we can now analyse a biological system

in depth by integrating different levels, such as ecology, physiology, genetics and cellular biology, using geospatial images². In the field of biological evolution, this has greatly helped us to better understand the relationships between species across different speciation events and their patterns.

I.A.2. Vertebrates' evolutionary history

The first common character of all vertebrates, and which start their tree, is the presence of a vertebrate column. We first found the jawless fish (which means “fish without jaws”). After there is the apparition of the jaws, with the cartilaginous fish. Then we found the skeleton with bones (with the bony fish), the four limbs and the lungs (which marks a life outside water, with the amphibians), then the presence of the amniotic egg (which had an important evolutive interest by the protection of the embryo, or the foetus, outside of an aquatic environment). Mammals are principally differentiated from other amniotic animals by the hair (Figure 1).

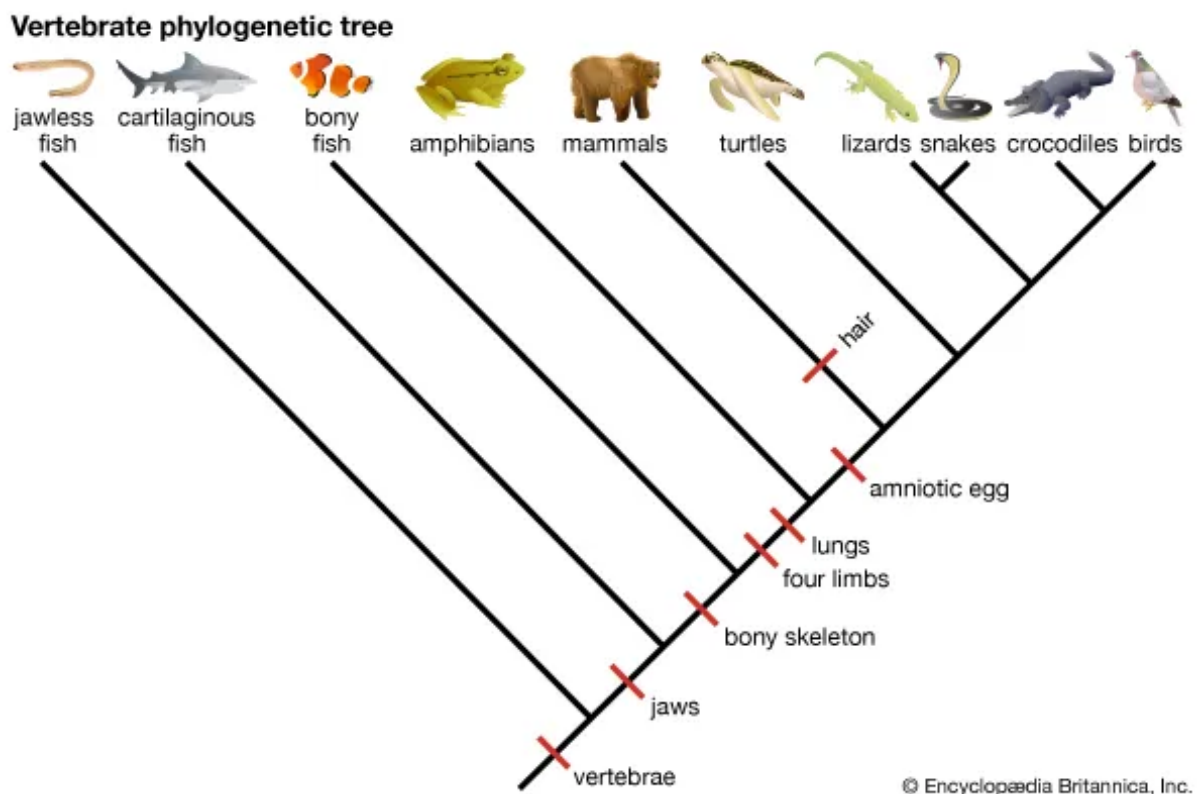


Figure 1: Vertebrate phylogenetic tree

Each embranchment shows the biggest evolution that permit to separate them into different group. For example: cartilaginous fish are vertebrate with jaws, but as they don't have a bony skeleton, they precede the vertebrate that had one, like bony fish, amphibians or mammals.

I.A.3. Group of interest for the study: bony fish

Bony fish (Osteichthyes) represent approximately half of all vertebrate species and exhibit remarkable phenotypic diversity, ranging from deep-sea anglerfish to surface-dwelling

salmon, and from elongated eels to compressed flatfish (Figure 2). This extraordinary diversity reflects their adaptation to virtually every aquatic niche across marine, freshwater, and brackish environments.

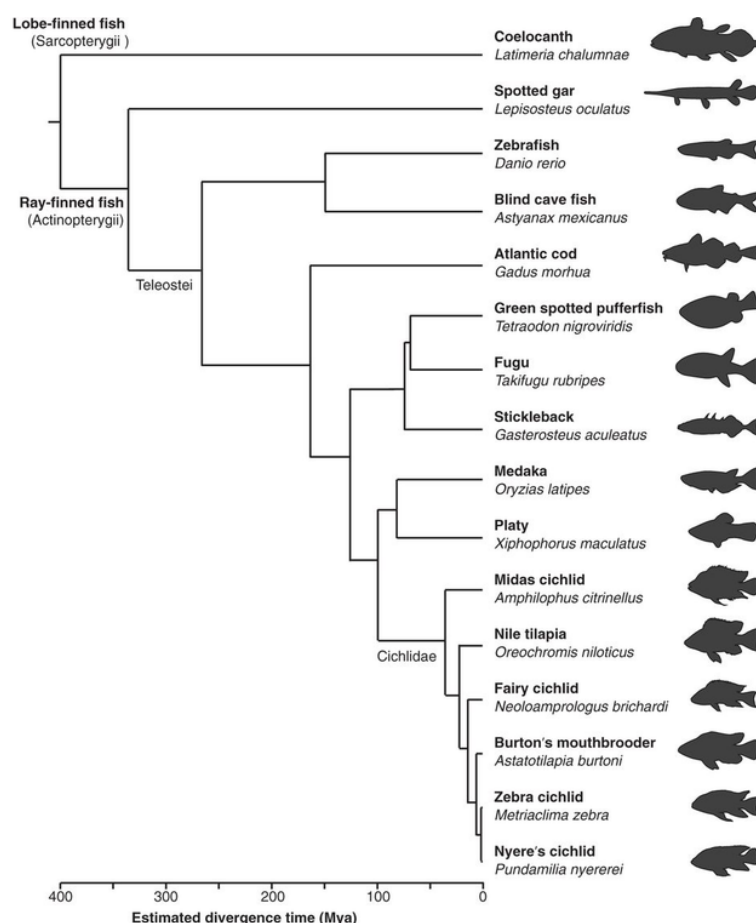


Figure 2 - Phylogeny of common bony fish species

Phylogenetic relationships among major bony fish lineages showing evolutionary divergence times and morphological diversity. The tree illustrates the deep evolutionary splits between major teleost groups (Lobe-finned fish, Ray-finned fish) and their subsequent radiation into diverse ecological niches. Estimated divergence times are shown in millions of years ago (Mya), highlighting the ancient origins of major lineages and the relatively recent diversification within teleost groups. Fish body represent the remarkable morphological diversity achieved through adaptive evolution across different aquatic environments.

Their evolutionary success stems partly from a whole-genome duplication event (Ts3R) that occurred approximately 300 million years ago in the teleost lineage, providing raw genetic material for adaptive diversification³. Unlike terrestrial vertebrates that face gradual environmental gradients, fish encounter dramatic pressure variations with depth—pressure doubles every 10 meters underwater, creating extreme selective pressures absent in terrestrial environments. Fish reproductive strategies are equally diverse, encompassing oviparity, viviparity, and unique forms like mouthbrooding and nest-building. Environmental parameters such as temperature, salinity, ionic composition, and hydrostatic pressure create complex selective landscapes that have driven the evolution of specialized physiological and morphological adaptations⁴.

Recent genomic sequencing initiatives have provided high-quality reference genomes spanning major fish lineages, enabling large-scale comparative genomics studies. This genomic foundation, combined with their phylogenetic position and experimental tractability in species like zebrafish and medaka, makes bony fish ideal models for understanding vertebrate evolution, gene family dynamics, and adaptive radiation processes.

B. Gene duplication

I.B.1. Principles

Gene duplication represents a fundamental evolutionary mechanism that has profoundly shaped the genomic architecture and phenotypic diversity of all living organisms. Gene duplication refers to the process by which genetic material is copied, resulting in multiple copies of genes, chromosomal segments, or entire genomes within an organism's DNA. This mechanism can occur through various molecular processes, ranging from local tandem duplications affecting single genes to large-scale events encompassing entire genomes.

The molecular causes of gene duplication are diverse and include unequal crossing over during meiosis, replication errors, transposable element activity, and polyploidization events⁵. Environmental pressures such as habitat changes, predation, or resource competition can favor the retention of duplicated genes that provide adaptive advantages. These duplications provide raw genetic material for evolutionary innovation, allowing one copy to maintain its original function while the other evolves new or modified functions⁶.

Gene duplications can occur at various stages of the cell cycle, with whole-genome duplications (WGDs) typically arising during meiotic or mitotic processes through errors in chromosome segregation. The timing of duplication events has been crucial in vertebrate evolution, with major WGD events occurring approximately 500-450 million years ago during early vertebrate evolution⁵, and additional events in specific lineages such as the teleost-specific duplication around 300 million years ago⁷.

Gene duplications affect both somatic and germline cells, but only those occurring in germline cells can be transmitted to offspring and thus contribute to evolutionary change. Gene duplication is a universal phenomenon across all domains of life, but its frequency and evolutionary impact vary significantly among different taxonomic groups⁶. Vertebrates have experienced multiple rounds of WGDs, with fish lineages showing particularly complex duplication histories.

Gene duplications enable phenotypic evolution and environmental adaptations through several mechanisms. Following duplication, gene copies can undergo sub functionalization, where ancestral functions are partitioned between duplicates, or neofunctionalization, where one copy evolves entirely new functions⁶. This evolutionary flexibility has been particularly important in vertebrate evolution, where WGDs have been associated with increased morphological complexity and adaptive radiation.

Vertebrates have experienced a complex history of WGDs relative to other animal lineages. The ancestral vertebrate genome underwent two rounds of WGD (1R and 2R) approximately 500-450 million years ago⁵, followed by a third round (3R or Ts3R) specific to teleost fish around 300 million years ago⁷. Some fish lineages experienced additional duplications, such as the salmonid-specific 4R (Ss4R) and the cyprinid-specific 4R (Cs4R)⁸, resulting in some species having up to eight copies compared to the ancestral single-copy genes found in invertebrates.

The relationships between genes following duplication events are classified using specific terminology. Homologous genes share common evolutionary origin and can be further subdivided into orthologs (genes in different species that evolved from a common ancestral gene through speciation events) and paralogs (genes within the same genome that arose through gene duplication)⁹. Ohnologs represent a special category of paralogs that specifically originated from WGD events, named after Susumu Ohno, who first proposed the importance of gene duplication in evolution¹⁰. These ohnologs often retain high levels of sequence similarity and frequently maintain coordinated expression patterns, indicating their recent evolutionary origin from genome-wide duplication events.

I.B.2. Importance in evolution: case of vertebrates

The evolutionary significance of gene duplication in vertebrates cannot be overstated, as these events have fundamentally shaped the genomic architecture and phenotypic complexity that characterizes this diverse clade. The timing and pattern of WGDs in vertebrate evolution reveal a striking correlation between duplication events and major evolutionary innovations that have defined vertebrate success across diverse ecological niches.

The two ancestral vertebrate WGDs (1R and 2R) (Figure 3) occurred during the Cambrian and Ordovician periods, coinciding with the emergence of complex body plans, sophisticated organ systems, and the evolution of jawed vertebrates⁵. These early duplications provided the genomic foundation for the evolution of elaborate developmental programs,

complex neural networks, and specialized physiological systems that distinguish vertebrates from their invertebrate relatives. The duplication of developmental regulatory genes, particularly transcription factors and signalling pathway components, enabled the evolution of more complex gene regulatory networks that could support increased morphological and physiological complexity⁶.

Evidence for the impact of these early WGDs can be observed in the structure of modern vertebrate genomes, where many gene families show characteristic patterns of four-fold paralogy reflecting the two successive duplication events⁵. The Hox gene clusters provide a paradigmatic example, with invertebrates typically possessing a single cluster while vertebrates maintain multiple clusters that have enabled the evolution of complex body segmentation and regional specialization along the anterior-posterior axis¹¹.

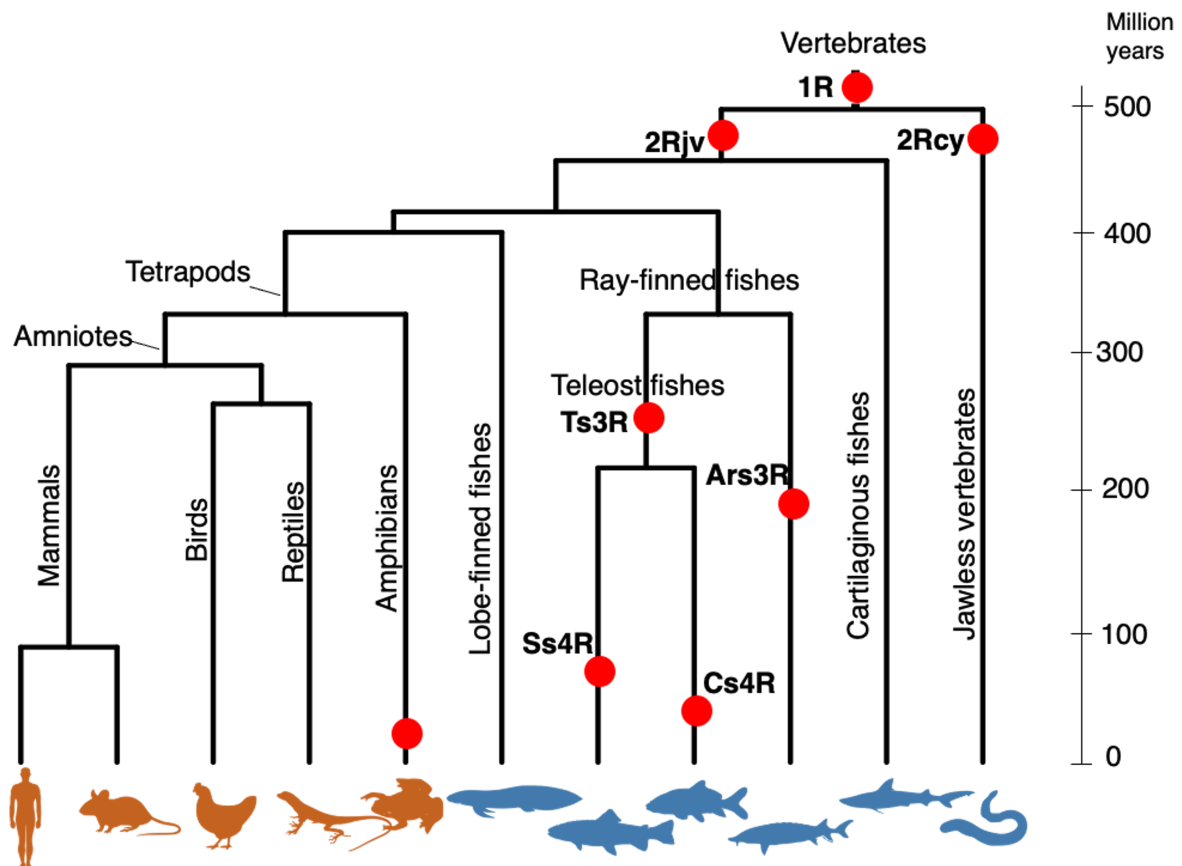


Figure 3 – Non-exhaustive vertebrates gene tree

Non-exhaustive vertebrate phylogenetic tree showing the distribution the species from major vertebrate lineages across evolution. The red dots on the phylogenetic tree indicate WGD events, revealing the complex duplication history that has shaped vertebrate genomes. The axis on the right shows the timescale in Million years.

The retention patterns of duplicated genes following WGDs are non-random and reveal important insights into the evolutionary forces acting on different gene categories. Genes involved in development, transcriptional regulation, and signal transduction show preferential

retention as duplicates, suggesting that dosage balance and the evolution of new regulatory interactions have been important drivers of duplicate gene maintenance¹². Conversely, genes involved in basic metabolic processes and housekeeping functions show higher rates of duplicate loss, likely reflecting the metabolic burden of maintaining redundant copies of essential functions.

I.B.3. Two different events in teleost

The evolutionary history of teleost fish is distinguished by a remarkable succession of WGD events that have fundamentally shaped their genomic architecture and contributed to their extraordinary diversification. Teleosts, which comprise approximately 96% of all fish species and nearly half of all vertebrate species, represent one of the most successful vertebrate radiations⁷, and their duplication history provides crucial insights into the relationship between genome evolution and phenotypic diversification.

The teleost-specific genome duplication (Ts3R) occurred approximately 300 million years ago, after the divergence of ray-finned fish from the lineage leading to land vertebrates but before the major teleost radiation⁷ (Figure 3). This event doubled the entire genome of the ancestral teleost, creating a situation where teleosts potentially possessed eight copies of genes that existed as single copies in the ancestral deuterostome. However, the reality is more complex, as extensive gene loss following the Ts3R has resulted in variable retention patterns across different gene families and lineages.

The Ts3R event has been particularly well-documented through comparative genomic analyses that reveal clear patterns of conserved synteny between teleost paralogous regions and their orthologous counterparts in other vertebrates⁷. Studies of model teleost species such as zebrafish, medaka, and pufferfish have demonstrated that approximately 15-20% of Ts3R duplicates have been retained as functional paralogs, with the remainder having been lost through pseudogenization or deletion events¹². The accelerated evolution and structural modifications of gene clusters following the Ts3R have been particularly evident in developmental gene families such as the Hox clusters¹³.

Following the Ts3R, specific teleost lineages experienced additional WGD events that further increased their genomic complexity. The salmonid-specific genome duplication (Ss4R) occurred approximately 80-100 million years ago in the lineage leading to salmon, trout, and their relatives⁸ (Figure 3). This event created an autotetraploid condition in salmonids, where individuals possess four copies of many genes compared to two copies in most other teleosts.

The Ss4R has been associated with the ecological success of salmonids in temperate and Arctic environments, potentially providing genetic flexibility for adaptation to extreme environmental conditions.

Similarly, the cyprinid-specific genome duplication (Cs4R) occurred in the lineage leading to carps and their relatives, creating another autotetraploid condition within teleosts⁸ (Figure 3). These lineage-specific duplications demonstrate that WGD continues to be an important evolutionary mechanism in fish evolution, contributing to the ongoing diversification of these highly successful vertebrate groups.

The retention patterns of genes following these fish-specific duplications reveal important insights into the evolutionary pressures acting on different functional categories. Genes involved in sensory perception, particularly vision and chemoreception, show elevated retention rates following fish-specific duplications, potentially reflecting the importance of these systems for survival in aquatic environments¹². Similarly, genes involved in metabolism and osmoregulation show preferential retention, suggesting that the physiological demands of aquatic life have favoured the maintenance of multiple gene copies in these functional categories.

The timing and pattern of gene loss following fish-specific duplications also reveal important evolutionary dynamics. Recent genomic analyses suggest that the rediploidization process following WGDs in teleosts has been relatively slow, with some duplicated chromosomal segments maintaining near-complete gene content for hundreds of millions of years⁸. This delayed rediploidization may have provided extended evolutionary opportunities for the evolution of new gene functions and regulatory interactions, contributing to the remarkable phenotypic diversity observed in modern teleost fishes.

C. Internship's context and objectives

I.C.1. Context

Recent publications of my internship supervisor and his team have focused on the development of comparative genomics resources, the creation of new databases and atlases that facilitate large-scale evolutionary analyses^{8,14,15}. Their research emphasizes the importance of automated pipelines for processing genomic data across multiple species, particularly in the context of understanding vertebrate evolution and the specific evolutionary trajectories of aquatic lineages.

The Ensembl database provides a tool called "Compara" which is a comprehensive comparative genomic resource¹⁶. When a new genome is added to the Ensembl database, they run it and publish a new version of these cross-species resources and analyses. For example, all Ensembl gene sequences are compared to one another to produce gene trees, infer homologues and produce gene families. This systematic approach enables researchers to track evolutionary relationships across diverse species and identify patterns of gene conservation, duplication, and loss that characterize different evolutionary lineages. However, while Ensembl Compara provides an excellent foundation for comparative genomics, its utility is primarily limited to species already integrated into the Ensembl database. This constraint significantly restricts researchers who wish to analyse non-model organisms or newly sequenced genomes that have not yet been incorporated into major genomic databases. Additionally, the Ensembl pipeline is designed for broad taxonomic coverage rather than focused analyses of specific evolutionary questions or targeted species groups.

Based on these online resources and its pipeline, a Galaxy workflow has been developed to find gene families¹⁷. The Galaxy implementation democratizes access to comparative genomics by providing a user-friendly web interface that eliminates the need for command-line expertise. They created Python scripts to adapt the different steps for Galaxy usage, making the complex Ensembl Compara methodology accessible to a broader research community. However, Galaxy workflows, while user-friendly, often lack the scalability and computational efficiency required for large-scale comparative genomics projects involving numerous species or extensive genomic datasets. My work is part of this continuation and represents a critical evolution in making comparative genomics tools more accessible and scalable.

I.C.2. Strategy and objectives

The primary objective is to develop a comprehensive automated pipeline that reconstructs the evolutionary history of vertebrate genes of a pool of chosen species and create an atlas, based on a Galaxy workflow¹⁷. With particular emphasis on fish species, which represent the core research focus of my hosting team. This pipeline will systematically identify gene families and trace their evolutionary trajectories through major evolutionary events, including gene duplication and speciation events that have shaped modern vertebrate genomes.

The pipeline will be fragmented into different sequential steps: downloading genomic data from NCBI or Ensembl databases, processing and standardizing these datasets across different annotation formats, creating gene families through homology detection and clustering algorithms, and reconstructing phylogenetic relationships within each family. A critical

component involves distinguishing between orthologous relationships arising from speciation events and paralogous relationships resulting from gene duplication events, as these distinctions are fundamental for understanding evolutionary processes.

The workflow incorporates advanced phylogenetic reconstruction methods that leverage both protein and nucleotide sequence information to maximize accuracy across different evolutionary timescales. For closely related species, nucleotide-based methods provide higher resolution, while protein-based approaches are more suitable for distant evolutionary relationships. The integration of synteny information through tools like *Scorpions*¹⁸ adds an additional layer of validation, ensuring that phylogenetic inferences are consistent with chromosomal organization and genomic context. Furthermore, the pipeline extends beyond gene family reconstruction to include ancestral genome inference through *Agora*¹⁹, enabling researchers to visualize and analyse the genomic composition of extinct ancestral species. This capability provides insights into ancient whole-genome duplications, lineage-specific gene expansions or contractions, and the temporal dynamics of genome evolution.

The final goal is to provide the research community with a user-friendly, reproducible, and scalable tool that democratizes access to comparative genomics analyses. By automating complex bioinformatics workflows and ensuring FAIR data principles, this pipeline will facilitate large-scale evolutionary studies and contribute to our understanding of vertebrate genome evolution, particularly in aquatic lineages where evolutionary pressures and adaptations present unique research opportunities.

II. Materials and methods

The development of the *EvoFish* pipeline required the integration of multiple computational tools and databases to create a comprehensive workflow for comparative genomics analysis. The methodology encompasses genome data acquisition from public repositories, systematic processing of heterogeneous annotation formats, phylogenetic reconstruction of gene families, and advanced evolutionary analysis including synteny-based tree correction and ancestral genome reconstruction. Each step of the pipeline was designed to maintain data integrity while ensuring scalability across diverse vertebrate species, with particular attention to the computational requirements of large-scale comparative studies. The workflow implementation prioritizes reproducibility through containerization and automated resource management, enabling consistent results across different computational environments.

A. Materials

The EvoFish pipeline was developed and implemented using Nextflow workflow management system (version 24.10.0) running on high-performance computing clusters with SLURM job scheduling. The complete pipeline source code, configuration files, and documentation are publicly available through the GitLab repository (https://gitlab.com/CorentinDumortier22/Evofish_Stage). Currently maintained as a private repository during development phase, the codebase will be made publicly accessible during summer with comprehensive documentation, installation guides, and citation guidelines.

Genomic data sources include the Ensembl database (release 95) for model organisms with high-quality annotations, and NCBI GenBank for additional species and recently sequenced genomes. The pipeline automatically retrieves genome assemblies, gene annotations in GFF3/GFF format, and coding sequences in FASTA format based on user-specified configuration files and species he wants.

Core computational tools integrated into the pipeline include EMBOSS transeq²⁰ v5.0.0 for sequence translation, Diamond²¹ most recent version for high-performance sequence alignment, hcluster_sg²² v0.5.1 for gene clustering, T-Coffee²³ v11.0.8 for multiple sequence alignment, and TreeBeST²⁴ v1.9.2 for phylogenetic reconstruction. Optional modules incorporate Scorpions¹⁸ lasted version for synteny-guided tree correction. We use Agora¹⁹ lasted version for ancestral genome reconstruction. All external dependencies are managed through Singularity containers and Conda environment to ensure reproducibility and consistent execution across different computing environments. The pipeline uses Python (version 3.11) for data processing scripts, with additional dependencies including BioPython for sequence manipulation, pandas for data analysis, and PyYAML for configuration management. System requirements include Linux-based operating systems, minimum 600GB RAM and 50CPUs for large-scale analyses, and multi-core processors for parallel execution. User will be able to choose the compute's parameters such as CPU, RAM or duration limit.

The results are based on the pipeline running of the following species : *H. sapiens*, *M. musculus*, *L. oculatus*, *D. rerio*, *O. mykiss*, *S. trutta*, *S. typhle*, *G. aculeatus* and *O. latipes* (Figure 4).

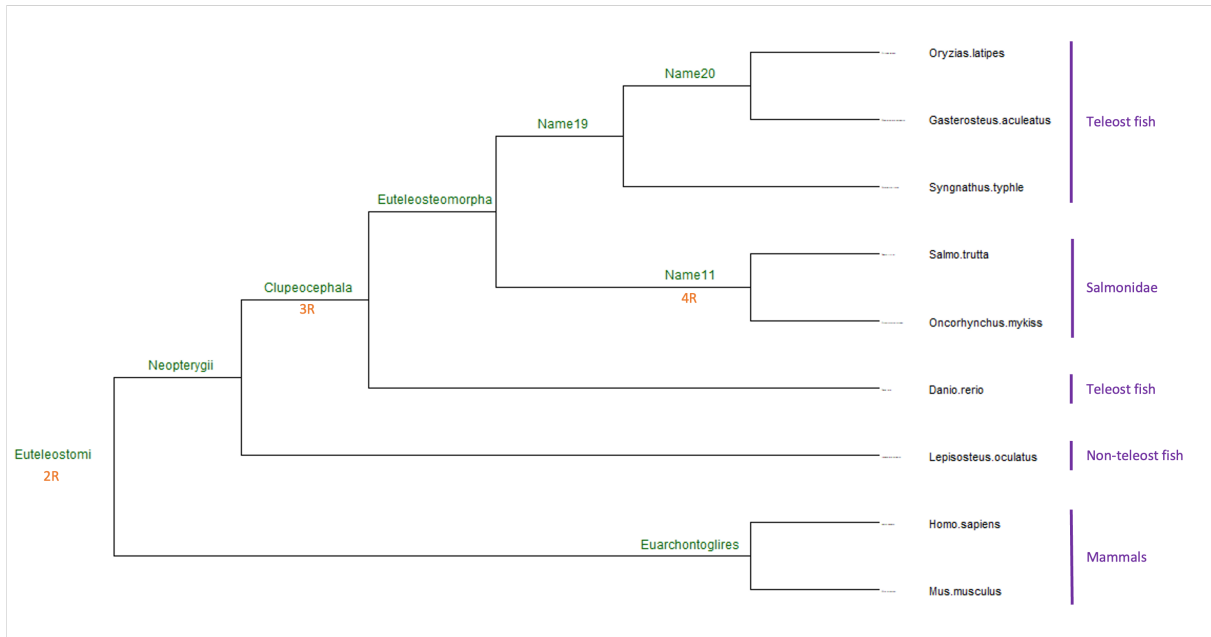


Figure 4 – Phylogenetic tree.
Phylogenetic tree of the nine species used to build the pipeline.

B. Methods

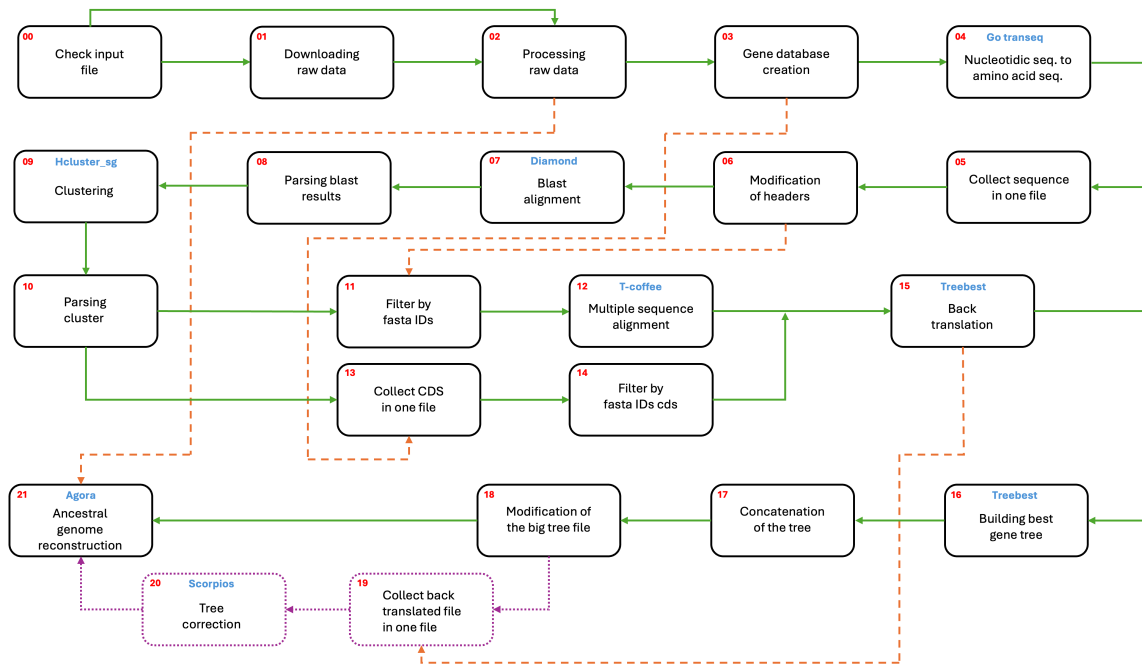


Figure 5 – Diagram of the pipeline workflow.

The pipeline processes genomic data from Ensembl and NCBI through 22 sequential modules, from input validation to ancestral genome reconstruction. The workflow begins with species-specific genome data download and processing (modules 00-03), followed by protein translation and homology detection (modules 04-08). Gene clustering and phylogenetic analysis are performed in modules 09-18, with an optional tree correction using Scorpios¹⁸ (module 20), and finish with ancestral genome reconstruction using Agora¹⁹ (module 21). Orange dashed lines indicate module interdependencies, and purple dotted lines indicate optional pathway.

II.B.1. From the input to clustering

Firstly, the pipeline looks at the validity of the configuration file the user gives. He must be composed of the species name, the assembly number, the Genomicus version and the Ensembl release (Figure 5 – step 00). Based on the assembly number and the Ensembl release, the pipeline downloads automatically the gene files (in `.gff3` format from Ensembl and `.gff` format from NCBI) and the transcript sequence (in `.cds` format) on Ensembl or NCBI database (Figure 5 – step 01). As the data from NCBI are not in the same format as Ensembl, we had to convert and modify them to be sure the gene database be created (Figure 5 – step 02). Then we created a gene database using the *gstf_preparation* python script (from the Galaxy tool Github²⁵) which generates an SQLite database containing gene features and produces filtered FASTA sequences for downstream analysis (Figure 5 – step 03).

The next step involves translating nucleotide sequences to protein sequences using transeq open-source tool from the EMBOSS suite²⁰ (Figure 5 – step 04). These protein sequences are then collected into a single file (Figure 5 – step 05) and their headers are modified to ensure compatibility with next tools (Figure 5 – step 06). To identify homologous relationships, we perform protein alignment using Diamond²¹, which conducts BLASTP searches of all sequences against themselves to identify potential orthologs and paralogs (Figure 5 – step 07). The resulting alignment scores are parsed and filtered to retain only significant hits, with weights calculated as minus log10 of the e-value divided by 2, capped at 100 (Figure 5 – step 08). Gene clustering is then performed using Hcluster_sg²², which groups sequences based on their similarity relationships into distinct gene families (Figure 5 – step 09). The clustering output is parsed to generate individual cluster files, with filtering to remove clusters containing fewer than 3 sequences, as this is the minimum required for meaningful phylogenetic analysis (Figure 5 – step 10).

II.B.2. From clustering to tree building

Each cluster is then processed to extract the corresponding protein sequences for multiple sequence alignment (Figure 5 – step 11). For each gene cluster identified in the previous step, multiple sequence alignment is performed using T-Coffee²³, which aligns the protein sequences within each family to identify conserved regions and evolutionary relationships (Figure 5 – step 12). Simultaneously, the pipeline collects all coding sequences (CDS) from the original genome annotations (Figure 5 – step 13) and filters them to match the gene clusters, ensuring that nucleotide sequences correspond to the aligned protein sequences (Figure 5 – step 14).

The protein alignments are then used to guide the back-translation of nucleotide sequences using TreeBeST backtrans²⁴, which maintains the alignment structure while preserving the original DNA sequences (Figure 5 – step 15). This step is crucial as phylogenetic reconstruction benefits from both protein-level conservation for distant relationships and nucleotide-level variation for recent evolutionary events. Gene tree construction is performed using TreeBeST best²⁴, which implements multiple phylogenetic methods and merges them into a consensus tree while minimizing duplications and deletions relative to a known species tree (Figure 5 – step 16). This approach leverages the fact that DNA-based methods are often more accurate for closely related sequences, while protein-based trees perform better at longer evolutionary distances. All individual gene trees are then concatenated into a single file (Figure 5 – step 17), and transcript identifiers are converted to gene identifiers to ensure consistency across the analysis (Figure 5 – step 18).

II.B.3. Synteny-guided tool for tree correction: Scorpions

We choose to put Scorpions as an option. If enabled, all back-translated alignments are collected to be used in synteny analysis (Figure 5 – step 19). Scorpions is a synteny-guided tool used to correct gene trees by identifying and resolving inconsistencies arising from duplication and speciation events¹⁸ (Figure 5 – step 20). Synteny, defined as the conservation of gene order and chromosomal organization between related species, provides crucial information for distinguishing true orthologous relationships from paralogous genes arising from duplication events (Figure 6).

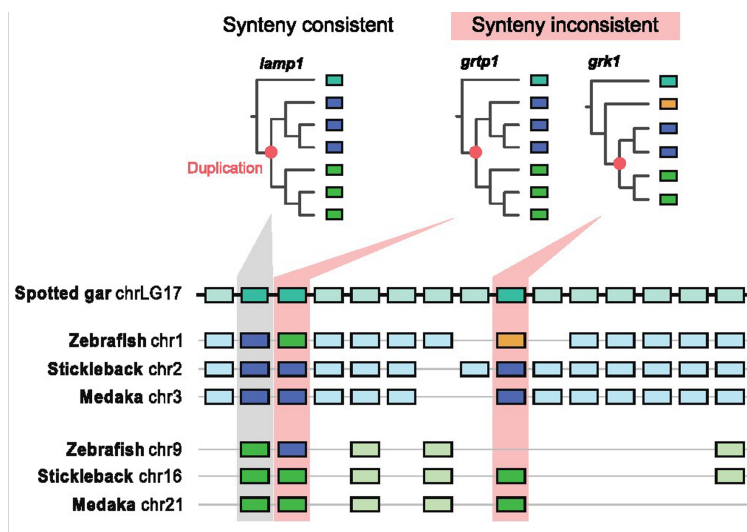


Figure 6 – Local synteny context is informative to sort out gene duplicates after WGD.

Gene trees and synteny context of the *lamp1*, *grtp1*, and *grk1* gene families in teleosts and their nonduplicated outgroup (spotted gar). Duplication nodes are labelled in red; blue and green boxes represent members of the two orthology groups in each original gene tree from Ensembl; yellow denotes genes annotated as nonduplicated. The synteny context identifies gene trees where zebrafish homologs are assigned to the wrong orthology groups (highlighted in red). Extracted from Scorpions article¹⁸.

Scorpions takes as input the concatenated gene trees from step 17, the species tree, genomic coordinates from GFF files, and the back-translated alignments collected in step 19. It uses this information to detect synteny breaks and gene order conservation across species, which helps distinguish between true orthologs and paralogs that may have been misclassified in the initial phylogenetic analysis. The tool applies synteny constraints to correct tree topologies, particularly focusing on whole-genome duplication events that can complicate orthology assignment. The output consists of corrected gene trees that better reflect the true evolutionary relationships between genes, considering both sequence similarity and chromosomal context.

II.B.4. Ancestral genome reconstruction: Agora

Agora is a computational tool used to reconstruct ancestral gene families and genomes based on comparative genomic data¹⁹ (Figure 5 – step 21). The tool employs a parsimony-based approach to infer the gene content and organization of extinct ancestral species by analysing the evolutionary relationships between extant genomes. The pipeline provides Agora with the corrected gene trees (either from Scorpions if enabled, or from the standard tree correction step), the species tree, and gene lists for each species derived from the genome annotations. Agora uses this information to reconstruct ancestral genomes by inferring gene gain and loss events along the phylogeny. The algorithm identifies conserved gene families across species and predicts which genes were present in common ancestors, as well as when specific gene duplications or losses occurred during evolution. The output includes reconstructed ancestral genomes with predicted gene content, chromosomal organization, and evolutionary scenarios explaining the transitions between ancestral and modern genomes. This information provides valuable insights into genome evolution, including the identification of ancient whole-genome duplications, lineage-specific gene expansions or contractions, and the evolutionary dynamics of gene families across the phylogeny.

III. Results

A. Data metrics

We choose our nine species (Figure 4) for specific reasons : i) they present different level of WGD 2R, 3R and 4R; ii) *H. sapiens* (human) and *M. musculus* (mouse) are among the most extensively studied vertebrate species; iii) fish reproduce in a wide variety of ways, for example, the female *O. latipes* (medaka) lays her eggs in a hole and the males come and deposit their sperm on top of them, involving a great deal of sperm competition, in contrast, for *S. typhle*

(pipefish), the female lays her eggs in a pouch in the male, who fertilises them inside, so there is almost no sperm competition.

Across the workflow, several filters are applied to keep the longest transcripts, the best alignment, best clusters and best gene tree. *S. trutta* has the biggest number of gene with 42405, where *L. oculatus* has the lowest value with 17238 genes (Figure 7A-B). For the transcripts, *H. sapiens* present the biggest value with 109095 transcripts, and *L. oculatus* has the lowest value with 22483 transcripts (Figure 7C-D). Between module 02 and module 16 the average loss of genes is 5%, with a maximum of 17% for *M. musculus* and a minimum of 1% for *L. oculatus* (Figure 7A-B). The average loss of transcripts is 52%, with a maximum of 82% for *H. sapiens* and a minimum of 24% for *L. oculatus* (Figure 7C-D).

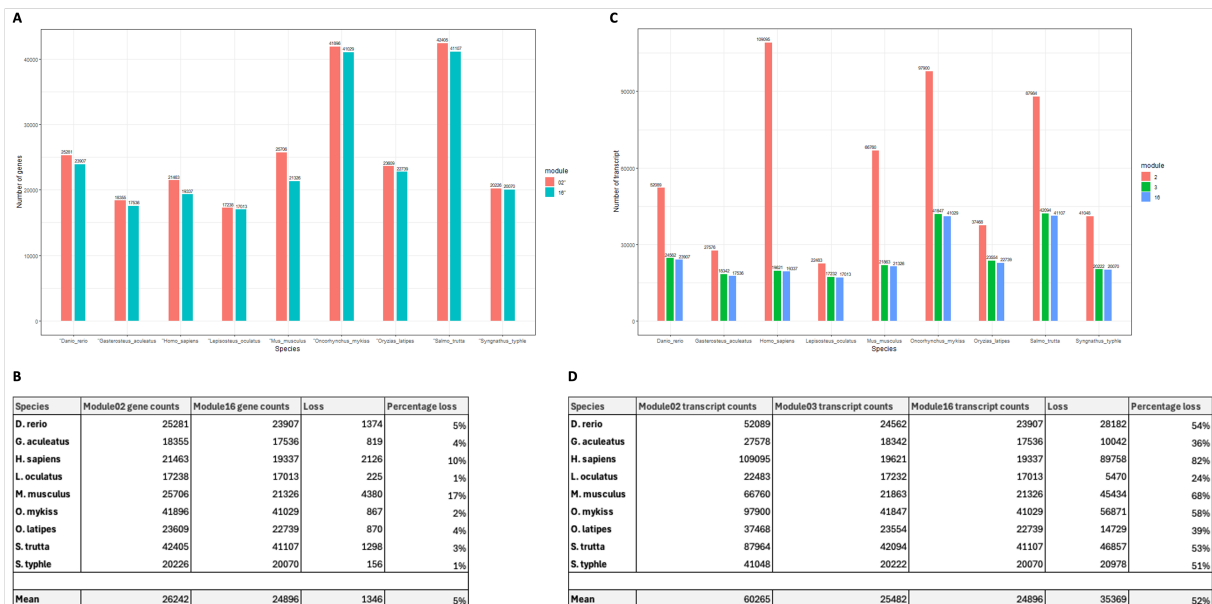


Figure 7 – Genes and transcripts metrics at different step.

(A) Number of genes by species, in red at module 02 (processing raw data), in blue at module 16 (TreeBeST best). (B) Number of transcripts by species, in red at module 02 (processing raw data), in green at module 03 (database creation), in blue at module 16 (TreeBeST best). (C) Summary table of genes metrics. (D) Summary table of transcripts metrics.

Trees correspond to gene family. After module 16 (TreeBeST best) we have 9090 trees (Table 1). They have different size, with an average gene per tree of 25, a minimum gene in tree of 2 and a maximum gene in tree of 1282. Also, some trees are composed of one specie, other with the 9 species studied, and an average species per tree of 6.8 (Table 1).

Table 1 – Genes metrics in trees.

	Total tree	Average genes per tree	Min genes in tree	Max genes in tree	Average species per tree	Min species in tree	Max species in tree
Values	9090	25	2	1282	6.8	1	9

B. Comparison tree with or without correction

We ran the pipeline twice. Once without the Scorpions option enabled, and once with it, to see how effective the correction was. Of the 9,090 trees, 1,683 were corrected, i.e. 18.5% of

them, with a total of 58,686 corrections, an average of 33 per tree, and a minimum of 5 and a maximum of 713 corrections observed (Table 2).

Table 2 – Tree correction made by Scorpions metrics.

	Corrected tree	Total correction	Percentage corrected tree	Average correction by tree	Min correction in tree	Max correction in tree
Values	1683	58686	18.5	33	5	713

To analyse the corrections made by Scorpions, we chose two different trees according to these criteria: i) presenting at least one species from each WGD rank in our study, a 2R individual, a 3R individual and a 4R individual, and between 15 and 20 genes; ii) also presenting all the species, but with a maximum of around ten genes.

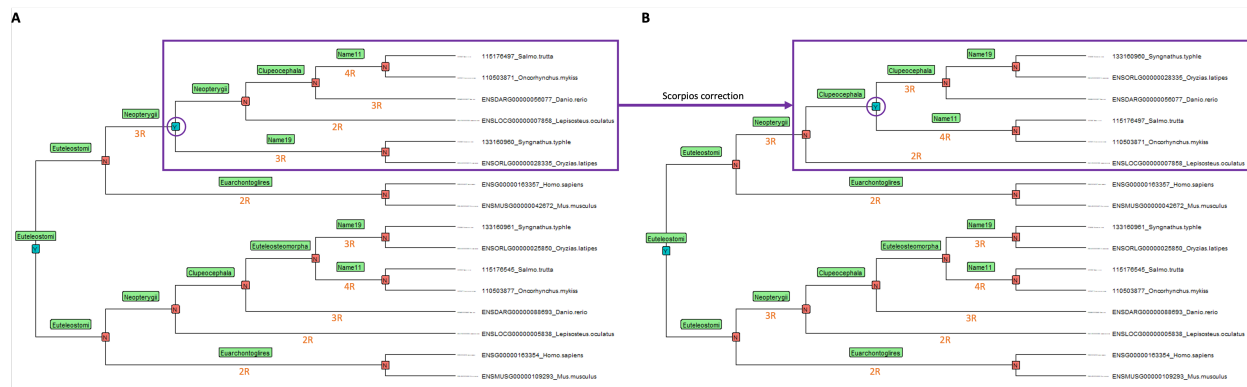


Figure 8 – Phylogenetic tree of a gene family, first example of synteny-guided correction.

Phylogenetic tree of a gene family, made up of 16 genes and 8 species with three 2R species *H. sapiens*, *M. musculus* and *L. oculatus*, three 3R species *D. rerio*, *S. typhle* and *O. latipes*, and two 4R species *O. mykiss* and *S. trutta*. (A) Tree without correction, directly taken after TreeBeST best step. (B) Tree with correction made by Scorpions. In green box the name of the node family. In blue box if it's a duplication event [Y]. In red box if it's a speciation event [N]. In orange the WGD rank : 2R for two events in species history, 3R for three events and 4R for four events. The purple square shows the part of the tree that has been corrected. The purple circle, the corrected node.

This tree contains the ENSG00000163354_Homo.sapiens gene (Figure 8), which is annotated as *DCST2* in the Ensembl database. The ENSMUSG00000109293_Mus.musculus gene (Figure 8) is annotated as *Dcst2*. The 115176545_Salmo.trutta and 110503877_Oncorhynchus.mykiss genes (Figure 8), referenced in the NCBI database, are annotated as 'dcst2 DC-STAMP domain containing 2'. Therefore, we can assume that it is the composite tree of the *DCST2* gene in *H. sapiens* and that they are all its orthologues.

The left tree (Figure 8A) begins with a node of duplication and two groups of paralogs genes from one ancestral gene. The bottom group follow correctly the phylogenetic species tree whereas the upper group shows an error according to the phylogenetic tree of species (Figure 4). The gene of the 2R species *L. oculatus* is placed on a higher branch than the gene of the 3R species *S. typhle* and *O. latipes* (Figure 8A - purple box). This is impossible because *L. oculatus* is a non-teleost 2R fish, and duplication occurs after speciation. Its gene cannot be classified as a post-3R gene. The right tree (Figure 8B) is its correction with the synteny-guided tool

Scorpios (Figure 6). We can see that the upper group of the tree has been corrected and the *L. oculatus*'s gene is now correctly placed, before the *Clupeocephala* ancestor, proposing a correct 2R/3R enchainment (Figure 8B – purple box), and becoming true according to the species tree (Figure 4). The duplication node has also been moved at *Clupeocephala* node, which marks the separation of 3R and 4R species. This correction demonstrates how synteny information can resolve phylogenetic inconsistencies that arise from artifacts in sequence-based tree reconstruction methods.

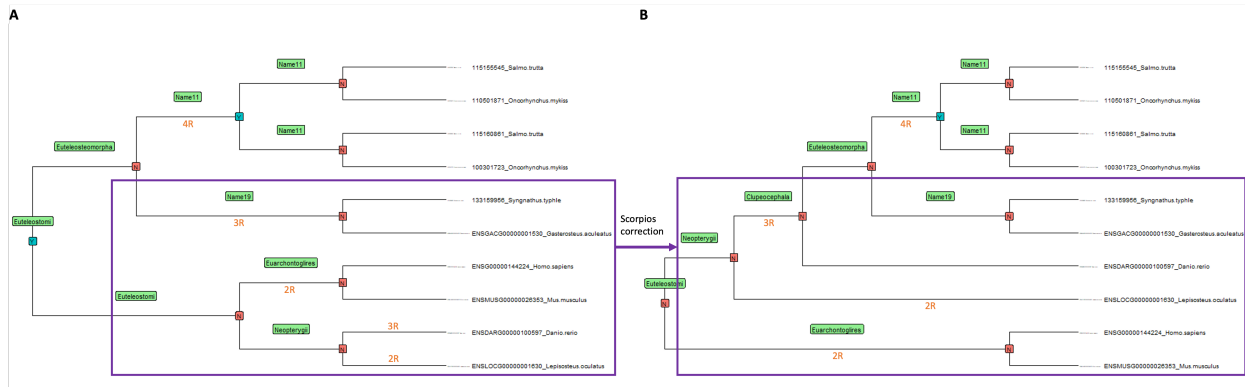


Figure 9 - Phylogenetic tree of a gene family, second example of synteny-guided correction.

Phylogenetic tree of a gene family, made up of 10 genes and 8 species with three 2R species *H. sapiens*, *M. musculus* and *L. oculatus*, three 3R species *D. rerio*, *S. typhle* and *G. aculeatus*, and two 4R species *O. mykiss* and *S. trutta*. (A) Tree without correction, directly taken after TreeBeST best step. (B) Tree with correction made by Scorpios. In green box the name of the node family. In blue box if it's a duplication event [Y]. In red box if it's a speciation event [N]. In orange the WGD rank : 2R for two events in species history, 3R for three events and 4R for four events. The purple square shows the part of the tree that has been corrected. The purple circle, the corrected node.

This tree contains the ENSG00000144224_Homo.sapiens gene (Figure 9), which is annotated as *UBXN4* in the Ensembl database. The ENSMUSG00000026353_Mus.musculus gene (Figure 9) is annotated as *Ubxn4*. The ENSGACG00000001530_Gasterosteus.aculeatus gene (Figure 9) is uncharacterised in Ensembl, whereas the ENSDARG00000100597_Danio.rerio gene (Figure 9) is annotated as *ubxn4*. Genes 115155545_Salmo trutta and 110501871_Oncorhynchus mykiss (Figure 9), referenced in the NCBI database, are annotated as 'UBX domain-containing protein 4-like'. Therefore, we can assume that it is the composite tree of the *UBXN4* gene in *H. sapiens* and that they are all its orthologues.

The top of the upper group on the left tree (Figure 9A) shows a correct pattern of duplication on 4R species *S. trutta* and *O. mykiss*. But there is an incoherence on the rest of the tree. On the bottom group, *D. rerio* is in the wrong place according to species tree (Figure 4), between the two mammal's species *H. sapiens* and *M. musculus* and non-teleost fish *L. oculatus* (which is also in the wrong side of the tree) (Figure 9A – purple box). The right tree (Figure

9B), that has been corrected with synteny-guided tool Scorpions, shows us the correct location of these two species, with correct ancestor *Neopterygii*, *Clupeocephala* and *Euarchontoglires* branch alone, becoming true according to the species tree (Figure 4). The duplication node has not moved and remains correct. Again, this tree correction demonstrates that synteny information are important to resolve phylogenetic inconsistencies.

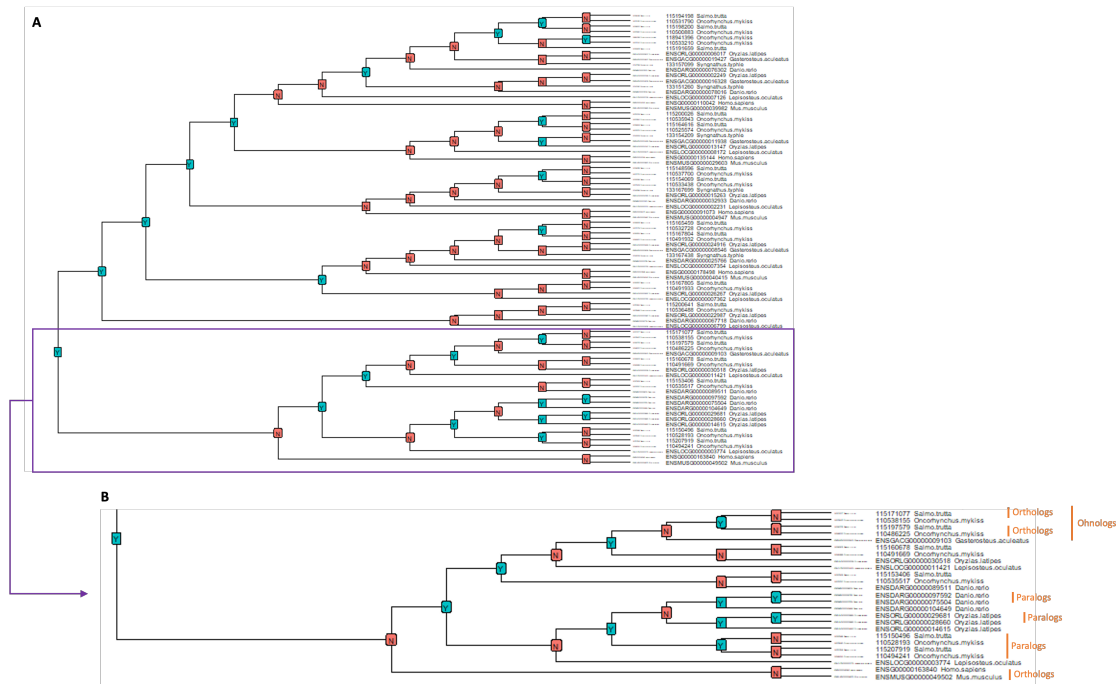


Figure 10 – Phylogenetic tree of a gene family, example of relationship between genes.

Phylogenetic tree of a gene family, made up of 83 genes and 9 species with three 2R species *H. sapiens*, *M. musculus* and *L. oculatus*, four 3R species *D. rerio*, *S. typhle*, *O. latipes* and *G. aculeatus*, and two 4R species *O. mykiss* and *S. trutta*. (A) The whole tree, corrected by Scorpions. (B) An enlargement of the lower branch of the tree. In purple box, the area enlarged. In orange, the relationship between two or more genes.

The trees reconstructed by the pipeline may be made up of many genes, which makes it difficult to analyse the relationships between them accurately (Figure 10A). Enlarging the lower branch allows us to look more closely at the relationships between genes in the same family in a smaller area (Figure 10B).

Nevertheless, we can put forward a hypothesis about the gene with which this family is associated. The ENSG00000163840_Homo.sapiens gene is annotated as *DTX3L* in the Ensembl database. The ENSMUSG00000049502_Mus.musculus gene is also annotated as *Dtx3l* on Ensembl. The three *D. rerio* genes ENSDARG00000089511, ENSDARG00000075504 and ENSDARG00000104649 are annotated as *dtx3la*, *dtx3lb.2* and *dtx3lb.3*, respectively. Finally, the two genes 115150496_Salmo.trutta and 110528193_Oncorhynchus.mykiss are both annotated as 'E3 ubiquitin-protein ligase DTX3L' in the NCBI database. Therefore, we can assume that it is the composite tree of the *DTX3L* gene in *H. sapiens* and that they are all its orthologues.

The *S. trutta* gene 115171077_Salmo.trutta and the *O. mykiss* gene 110538155_Oncorhynchus.mykiss, at the very top of the tree, are orthologous because they diverged through a speciation event when the Salmo and Oncorhynchus lineages separated, and their last common ancestor in the gene tree represents this speciation event rather than a gene duplication event (Figure 10B). They correspond to two uncharacterised genes. The other two genes just below, 115197579_Salmo.trutta and 110486225_Oncorhynchus.mykiss are also orthologous for the same reasons (Figure 10B). They correspond to a hypothetical protein for *S. trutta* gene and an uncharacterised gene for *O. mykiss*. These two groups of orthologous genes are ohnologous because they originated from the salmonid-specific whole genome duplication (4R WGD), creating duplicated copies of ancestral genes that were subsequently retained in both lineages after the Salmo-Oncorhynchus speciation (Figure 10B).

On the middle branches, we can see a case of paralogous genes. The *D. rerio* (zebrafish) genes ENSDARG00000097592_Danio.rerio and ENSDARG00000075504_Danio.rerio are paralogous because they are duplicated copies of the same ancestral gene within the same species that arose through a gene duplication event, as evidenced by their clustering together in the phylogenetic tree with a duplication node (Figure 10B). These two genes correspond respectively to the dtex3lb.1 and dtx3lb.2 genes. The same applies to the two genes ENSORLG00000029681_Oryzias.latipes and ENSORLG00000028660_Oryzias.latipes from the species *O. latipes* (Japanese medaka), which correspond to two unannotated genes.

Finally, at the bottom of the branch we have the case of two pairs of co-orthologous genes. The two genes 115150496_Salmo.trutta and 110528193_Oncorhynchus.mykiss are co-orthologous with the two genes 115207919_Salmo.trutta and 110494241_Oncorhynchus.mykiss (Figure 10B). This is an interesting case, because it indicates that an ancestral gene duplication event occurred before the speciation between Salmo and Oncorhynchus lineages, creating two paralogous copies of the gene that were then inherited by both species after their divergence, resulting in co-orthologous relationships where each species retained both duplicated copies.

C. Ancestral genome reconstruction and synteny visualisation

With the trees reconstructed by TreeBeST best, with or without the synteny-guided correction by Scorpions, we wanted to reconstruct the ancestral genomes with Agora. The analysis reveals species-specific patterns of CAR gene dispersal. 4R species show less fragmentation of ancestral chromosomal regions than 2R/3R species (Figures 11A-B).

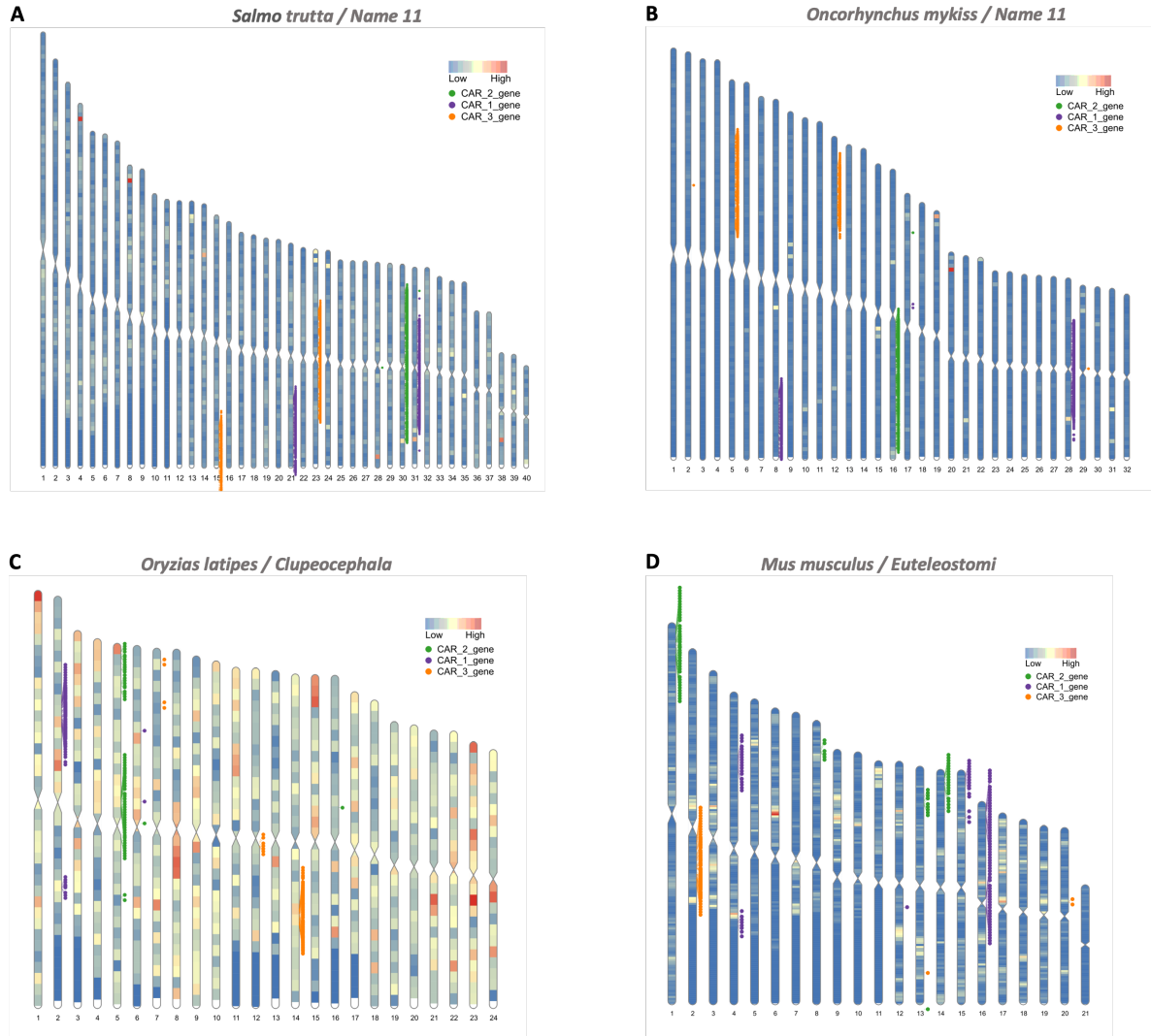


Figure 11 – Comparative synteny analysis of ancestral chromosome regions (CARs) across four vertebrate species

Chromosome ideograms showing the distribution of three ancestral chromosome regions (CAR_1, CAR_2, CAR_3) across four vertebrate species with different whole genome duplication histories. (A) *Salmo trutta* - 4R species with 40 chromosomes, painted with the three ancestral chromosome regions of Name11 ancestor. (B) *Oncorhynchus mykiss* - 4R species with 29 chromosomes, painted with the three ancestral chromosome regions of Name11 ancestor. (C) *Oryzias latipes* - 3R species with 24 chromosomes, painted with the three ancestral chromosome regions of Clupeocephala. (D) *Mus musculus* - 2R species with 19 chromosomes, painted with the three ancestral chromosome regions of Euteleostomi ancestor. Each chromosome is represented as a vertical bar with gene density in a one million base-paired window displayed as a blue heatmap (low to high intensity gradient). Coloured circles, linked to their position with line, indicate the chromosomal positions of genes belonging to each ancestral chromosome region: purple circles for CAR_1 genes, green circles for CAR_2 genes and orange circles for CAR_3 genes.

In the two 4R species, *S. trutta* and *O. mykiss*, CAR_1 and CAR_3 are fragmented across two different modern chromosomes: CAR_1 is fragmented across chromosomes 21–31 in *S. trutta* and 8–28 in *O. mykiss*, and CAR_3 is fragmented across chromosomes 15–23 in *S. trutta* and 5–12 in *O. mykiss* (Figure 11A–B). CAR_2 is highly conserved in a single block on chromosomes 30 and 16 for *S. trutta* and *O. mykiss*, respectively (Figure 11A–B). Of the 4R species, *O. mykiss* exhibits slightly greater fragmentation than *S. trutta*, which may reflect lineage-specific variations in chromosomal stability.

For the 3R species *D. rerio*, the fragmentation of the different ancestral CAR chromosomal regions is more pronounced. CAR_1 is divided into two parts: a large part on the upper part of chromosome 2 and a small part on the lower part (Figure 11C). There are also two small parts on chromosome 6. A similar pattern is observed for CAR_2: a large part is in the middle of chromosome 6, a smaller part is located at the beginning, and a few blocks are located elsewhere (Figure 11C). CAR_3 is fragmented across three chromosomes: a large part is found on chromosome 14, a small part on chromosome 12, and several blocks on chromosome 7 (Figure 11C).

Finally, for the 2R *M. musculus* species, fragmentation is much more pronounced. CAR_1 is fragmented into five sub-parts on four different chromosomes: 4, 12, 15 and 16 (Figure 11D). CAR_2 is also fragmented across four different chromosomes: 1, 8, 13 and 14 (Figure 11D). CAR_3 is concentrated on chromosome 2 (Figure 11D).

The progressive increase in fragmentation from 4R to 2R species reflects the cumulative effect of chromosomal rearrangements over evolutionary time.

IV. Discussion

The EvoFish pipeline successfully processed nine fish species, generating 9090 gene trees through automated comparative genomics analysis. The substantial filtering of genes and transcripts throughout the pipeline execution (52% average transcript loss) reflects the stringent quality control measures necessary for reliable phylogenetic inference. Species-specific variation in filtering rates, ranging from 24% in *L. oculatus* to 82% in *H. sapiens*, likely reflects differences in genome annotation quality and completeness across the studied taxa.

The correction of 1683 gene trees (18.5%) by SCORPiOs demonstrates the substantial impact of synteny-guided approaches on phylogenetic accuracy. With 58686 total corrections averaging 33 per tree, these results confirm that sequence-based phylogenetic methods alone are insufficient for resolving evolutionary relationships in post-whole-genome duplication lineages. This correction rate aligns with previous estimates suggesting that approximately one in four teleost gene trees contains topological inconsistencies when based solely on sequence evolution. The phylogenetic examples presented illustrate the power of synteny information to resolve complex evolutionary relationships. In the first corrected tree example (Figure 8), SCORPiOs successfully repositioned duplication nodes to reflect the known whole-genome duplication events affecting different lineages. The distinction between 2R species (*H. sapiens*, *M. musculus*, *L. oculatus*), 3R species (*D. rerio*, *S. typhle*, *O. latipes*), and 4R species (*O.*

mykiss, *S. trutta*) becomes clearly defined after correction, providing accurate evolutionary context for gene family evolution. The corrected gene trees enable precise identification of orthologous and paralogous relationships that are critical for functional genomics studies (Figure 10). The examples demonstrate clear cases of orthologous genes arising from speciation events, such as the *S. trutta* and *O. mykiss* gene pairs that diverged following the *Salmo*-*Oncorhynchus* split. These relationships are particularly important for understanding species-specific adaptations and the evolutionary fate of duplicated genes.

The identification of orthologous relationships resulting from the salmonid-specific 4R WGD provides insights into long-term duplicate gene retention. The maintenance of both gene copies in *S. trutta* and *O. mykiss* lineages suggests that these duplicates may have acquired distinct functional roles through subfunctionalization or neofunctionalization processes. Such patterns are essential for understanding how whole-genome duplications contribute to evolutionary innovation and phenotypic diversification in teleost fishes. Paralogous relationships within species, exemplified by the *D. rerio* *dtex3lb.1* and *dtx3lb.2* genes, demonstrate the pipeline's ability to correctly identify intraspecific duplications. The co-orthologous relationships observed in the bottom clade illustrate more complex scenarios where ancestral duplications preceded speciation events, creating evolutionary constraints that influenced subsequent lineage evolution.

The integration of Agora for ancestral genome reconstruction provides insights into the evolutionary history of vertebrate genomes. The comparative analysis of ancestral chromosome regions (CARs) across species with different whole-genome duplication histories reveals distinct patterns of chromosomal reorganization following major evolutionary events. The differential fragmentation observed between 2R, 3R, and 4R species demonstrates that genome evolution following whole-genome duplications follows predictable patterns related to the number of duplication events experienced by each lineage.

The 4R salmonid species (*S. trutta* and *O. mykiss*) exhibit relatively conserved synteny patterns, with most ancestral chromosome regions maintained in large blocks across few chromosomes. This pattern suggests that the recent salmonid-specific whole-genome duplication has had insufficient time for extensive chromosomal rearrangements, consistent with the delayed rediploidization hypothesis proposed for recent polyploidy events¹⁹. The concentration of CAR_2 genes in single chromosomal blocks in both salmonid species indicates strong selective constraints maintaining chromosomal integrity for specific ancestral regions.

In contrast, the 3R species *O. latipes* shows intermediate levels of chromosomal reorganization, with ancestral regions fragmented across multiple chromosomes but still maintaining recognizable syntenic blocks. This pattern reflects the longer evolutionary time since the teleost-specific whole-genome duplication, allowing for gradual chromosomal rearrangements while preserving essential syntenic relationships. The 2R species *M. musculus* displays the most extensive fragmentation, with ancestral chromosome regions distributed across multiple modern chromosomes, consistent with the ancient timing of the vertebrate whole-genome duplications and subsequent extensive chromosomal evolution.

These ancestral genome reconstructions provide critical temporal context for understanding the evolutionary dynamics of gene families and chromosomal organization. The ability to trace the evolutionary fate of specific chromosomal regions across major vertebrate lineages offers insights into the constraints and opportunities that have shaped modern genome architectures. The observation that certain ancestral regions remain more intact than others suggest differential selective pressures on chromosomal organization, potentially reflecting functional constraints related to gene regulation and chromosomal domains.

However, analysing only three ancestral chromosome regions (CAR_1, CAR_2 and CAR_3) provides a limited view of the full ancestral genome architecture. To accurately quantify the extent of post-WGD genome reorganization in different vertebrate lineages, a comprehensive assessment of rediploidization patterns and chromosome rearrangement rates would require examination of all ancestral chromosome regions. In the case of *S. trutta* and *O. mykiss*, a comprehensive analysis of CARs, including other salmonids, would enable the location of the Name 11 ancestor with greater certainty, either before or after the duplication event. It would also be interesting to include cyprinids, which have undergone a different fourth duplication event to salmonids⁸, in the study.

The Nextflow implementation addresses critical scalability and reproducibility challenges in comparative genomics. Unlike existing Galaxy-based workflows that prioritize user accessibility, the containerized approach ensures consistent execution across diverse computational environments while maintaining full reproducibility¹⁷. The modular pipeline architecture enables researchers to incorporate newly sequenced genomes without extensive reconfiguration, addressing limitations of database-dependent resources like Ensembl Compara¹⁶. The integration of established tools (TreeBeST, SCORPiOs, Agora) within a unified framework reduces technical barriers that have historically limited access to sophisticated evolutionary analyses.

This democratization of computational resources is particularly valuable for research groups studying non-model organisms or recently sequenced genomes not yet integrated into major genomic databases. The pipeline's adherence to FAIR data principles ensures that results can be effectively shared and reused by the broader research community, facilitating large-scale comparative studies across diverse teleost lineages.

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Résumé

Les téléostéens représentent l'un des groupes de vertébrés les plus diversifiés, avec environ la moitié des espèces vertébrées connues. Cette diversité remarquable résulte en partie de duplications génomiques complètes (WGD) qui ont fourni le matériel génétique nécessaire aux innovations évolutives. Le pipeline EvoFish développé lors de ce stage automatise l'analyse génomique comparative d'espèces souhaitées, représentant différents niveaux de duplication génomique (2R, 3R, 4R). Implémenté sous Nextflow pour assurer la reproductibilité et l'évolutivité, ce pipeline intègre des outils établis (TreeBeST, SCORPiOs, Agora) dans un workflow unifié de 22 modules. Il traite automatiquement les données génomiques depuis les bases Ensembl et NCBI, effectue la reconstruction phylogénétique des familles de gènes, applique des corrections guidées par la synténie et permet la reconstruction de génomes ancestraux. Les résultats montrent que 9090 arbres géniques ont été générés, dont 18,5% ont été corrigés par SCORPiOs, démontrant l'importance des approches guidées par la synténie pour résoudre les relations évolutives post-duplication. Les analyses de génomes ancestraux révèlent des patterns différentiels de réorganisation chromosomique entre les lignées, avec une fragmentation progressive des régions ancestrales selon le temps évolutif. Ce pipeline démocratise l'accès aux analyses génomiques comparatives en respectant les principes FAIR, facilitant l'étude de l'évolution des génomes de vertébrés, particulièrement dans les lignées aquatiques où les pressions évolutives créent des opportunités d'adaptation uniques.

Mots clés : Génomique évolutive ; Duplication génomique ; Téléostéens ; Phylogénie ; Pipeline

Abstract

Teleost fish represent one of the most diversified vertebrate groups, comprising approximately half of all known vertebrate species. This remarkable diversity stems partly from whole-genome duplications (WGDs) that provided genetic material for evolutionary innovations. The EvoFish pipeline developed during this internship automates comparative genomic analysis of specific species representing different levels of genome duplication (2R, 3R, 4R). Implemented using Nextflow to ensure reproducibility and scalability, this pipeline integrates established tools (TreeBeST, SCORPiOs, Agora) within a unified 22-module workflow. It automatically processes genomic data from Ensembl and NCBI databases, performs phylogenetic reconstruction of gene families, applies synteny-guided corrections, and enables ancestral genome reconstruction. Results demonstrate that 9,090 gene trees were generated, with 18.5% corrected by SCORPiOs, highlighting the importance of synteny-guided approaches for resolving post-duplication evolutionary relationships. Ancestral genome analyses reveal differential patterns of chromosomal reorganization across lineages, with progressive fragmentation of ancestral regions correlating with evolutionary time. This pipeline democratizes access to comparative genomics analyses while adhering to FAIR principles, facilitating studies of vertebrate genome evolution, particularly in aquatic lineages where evolutionary pressures create unique adaptation opportunities. The automated workflow addresses scalability and reproducibility challenges in comparative genomics, enabling researchers to incorporate newly sequenced genomes without extensive reconfiguration. By providing a user-friendly, containerized solution, EvoFish contributes to understanding the evolutionary dynamics of gene families and chromosomal organization across vertebrate lineages.

Keywords: Evolutionary genomics; Genome duplication; Teleosts; Phylogeny; Pipeline